

Metriq-gym: Unified benchmarking framework for quantum computers

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Abstract—As quantum hardware continues to advance across a growing landscape of vendors, it is increasingly important to evaluate device capabilities for specific computational tasks. However, cross-platform benchmarking remains challenging due to diverse hardware architectures, proprietary access interfaces, and the lack of standardized benchmarking frameworks. We introduce *metriq-gym*, an open-source Python framework designed to facilitate both system-level and application-level benchmarking of quantum devices across multiple providers. The *metriq-gym* framework offers a unified workflow for executing a suite of community-driven benchmarks, enabling consistent comparisons across heterogeneous quantum platforms. In this poster, we provide an overview of the *metriq-gym* architecture and demonstrate its capabilities through a curated set of benchmark results obtained from several leading quantum hardware systems.

I. INTRODUCTION

Over the past decades, quantum computing has transitioned from theoretical promise to experimental realization, with increasing qubit counts, improved fidelities, and broader access to quantum hardware. As this ecosystem matures, the need to reliably evaluate, compare, and track performance across devices has become urgent. Benchmarks not only guide hardware development, but also help application researchers determine which devices are suitable for their specific computational needs. They can capture the interplay between critical properties such as circuit quality, computational speed, and system scalability [1–4].

Despite this need, benchmarking in the current quantum landscape remains fragmented. Each hardware provider employs different qubit modalities, gate sets, error mitigation techniques, and software interfaces, making it challenging to conduct standardized assessments. Existing benchmarking tools [3–5] often cater to a single provider or are optimized for internal performance validation rather than open, reproducible,

cross-platform comparisons. Furthermore, benchmarks are frequently defined with limited transparency and may not reflect community consensus or real-world application relevance.

The absence of a community-driven, modular benchmarking framework that spans providers has hindered the ability to make meaningful and fair comparisons. This gap in the benchmarking infrastructure stifles innovation, obscures device capabilities, and complicates the development of performance metrics tied to practical quantum applications. In this work, we introduce *metriq-gym* that addresses these limitations by offering a modular, extensible, and community-curated benchmarking suite.

II. FRAMEWORK OVERVIEW

metriq-gym is an open-source Python package [6] that serves as an extensible benchmark framework for quantum computers. *metriq-gym* allow researchers, developers, and hardware evaluators to define, dispatch, and analyze benchmark jobs across multiple quantum hardware providers using a single, unified, and vendor-agnostic interface. In its current version, the framework offers the following features:

- a programming interface, so that new cross-platform benchmarks can be easily written in a way that is agnostic to the hardware-specific programming API.
- a command-line interface (CLI) that abstracts backend-specific operations such as transpilation, credential management, and device selection.
- a mechanism for specifying and validating benchmark parameters via JSON schema, ensuring parameter transparency, reproducibility, and community review.
- a mechanism for testing benchmark end-to-end on offline simulators before running them on real hardware.

The benchmark results can be uploaded to metriq.info, a companion platform for aggregation, visualization, and public access of results [7].

metriq-gym supports a growing list of quantum devices across several cloud platforms, including IBM Cloud, IonQ,

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AWS Braket, Azure Quantum, and devices integrated by leveraging qBraid-SDK, with users adding their credentials to access the providers. Its modular design allows seamless integration of new benchmarks and hardware backends, while its open development model fosters community participation in shaping benchmark definitions, refining metrics, and validating implementation correctness.

By promoting openness, interoperability, and standardization, *metriq-gym* aims to become a benchmarking foundation for the quantum computing field, supporting diverse use cases ranging from hardware validation to application performance forecasting.

III. EXAMPLES OF IMPLEMENTED BENCHMARKS

A. System-level benchmarking

We implemented key system-level metrics including:

- Quantum Volume (QV) [8]: Assesses the ability to execute random quantum circuits with high fidelity.
- Circuit Layer Operations Per Second (CLOPS) [9]: Measures execution speed of quantum gate layers.
- Bell State Effective Qubits (BSEQ): Evaluates entanglement capability via CHSH inequality violation.

Benchmarking results highlight platform-specific trade-offs—e.g., higher QV on one provider or architecture vs. faster CLOPS on another—emphasizing the need for diverse metrics to capture different hardware strengths.

B. Application-level benchmarking

To bridge system metrics with real-world utility, *metriq-gym* also supports application-level benchmarking such as:

- Quantum Machine Learning (QML) Kernel Benchmarking [10]: Measures the device’s ability to accurately compute kernel matrix elements using ZZ feature maps.
- Linear-Ramp QAOA Benchmarking: This benchmark tests quantum devices using a linearly interpolated version of the Quantum Approximate Optimization Algorithm, which offers simpler parameter schedules. It assesses a device’s ability to solve combinatorial optimization problems.

metriq-gym is also being extended to include standardized application benchmarks from the QED-C suite [3, 4], such as the Bernstein–Vazirani algorithm and Quantum Fourier Transform. These additions further broaden the scope of *metriq-gym* and align it with community efforts to define representative workloads for near-term quantum devices.

IV. DISCUSSIONS AND OUTLOOK

In summary, the *metriq-gym* framework provides a unified, extensible, and open-source approach to benchmarking quantum computers across diverse hardware platforms. By enabling consistent evaluations using a community-curated suite of benchmarks, *metriq-gym* addresses critical challenges in the quantum ecosystem, including fragmented benchmarking tools, lack of transparency, and platform-specific limitations. It

simplifies the process of running and comparing benchmarks on systems from providers such as IBM, IonQ, AWS, Azure, and others integrated through qBraid, thereby facilitating fair and reproducible assessments.

While *metriq-gym* significantly lowers the barrier to standardized benchmarking, several challenges persist. These include hardware-specific optimizations that may introduce performance biases, limited backend transparency regarding error mitigation and calibration procedures, and the inherent stochasticity of quantum systems, which affects result reproducibility. Addressing these issues will require continued community collaboration and methodological refinement.

Looking ahead, *metriq-gym* will expand to include new benchmarking modalities, such as analog quantum devices, extended application-level tests, and logical qubit benchmark.

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